Cloud Computing
LAB-6

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Section: F

1. Pulling Jenkins

`docker pull jenkins/jenkins:lts`

2. Building Jenkins images

`docker build -t jenkins-docker -f Dockerfile.jenkins .`

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$ docker bu
ilder prune -f
ID                                              RECLAIMABLE     SIZE            LAST ACCESSE
D
cmoh8hp2f9kklarx6150xein7*                      true            4.096kB         13 minutes a
go
d0fnix1pptcyca5dm2gw9s1oy*                      true    8.192kB         13 minutes ago
wr6r8fhymluowtvu81o1w0jkd*                      true    8.192kB         About a minute ago
xdqtj68nc8rd2q2tgw3jam7gy*                      true    4.096kB         About a minute ago
Total:  24.58kB
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$ docker bu
ild -t jenkins-docker -f Dockerfile.jenkins .
[+] Building 1.2s (5/5) FINISHED                                          docker:default
 => [internal] load build definition from Dockerfile.jenkins                          0.1s
 => => transferring dockerfile: 92B                                                   0.0s
 => [internal] load metadata for docker.io/jenkins/jenkins:lts                        0.1s
 => [internal] load .dockerignore                                                     0.1s
 => => transferring context: 2B                                                       0.0s
 => [1/1] FROM docker.io/jenkins/jenkins:lts@sha256:23b84d1066dbe69f2c17288c83ac07112  0.3s
 => => resolve docker.io/jenkins/jenkins:lts@sha256:23b84d1066dbe69f2c17288c83ac07112  0.0s
 => exporting to image                                                                0.3s
 => => exporting layers                                                               0.0s
 => => exporting manifest sha256:048cde9855d88132920eef112549cc31ba28b739cd715101a231  0.0s
 => => exporting config sha256:32f868ba857e9ac05ae98ad7c5413907d364852b20e8e04a450da5  0.0s
 => => exporting attestation manifest sha256:1d9ca8bd2b850a91feaafa6bcf82a8ee6bb986e9  0.1s
 => => exporting manifest list sha256:76b9b2e8bc9d0ef62d498de06e81a2a59aa0e56fa506172  0.0s
 => => naming to docker.io/library/jenkins-docker:latest                              0.0s
 => => unpacking to docker.io/library/jenkins-docker:latest                           0.0s
```

3. Run Jenkins container

` docker run -d -p 8080:8080 -p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
--name jenkins \
jenkins-docker
`

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$ docker ru
n -d -p 8080:8080 -p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
--name jenkins \
jenkins-docker
c795e0e45c827816568c067476a099d0ddf4e9f101777bfb255e816f2b98e9a8
```

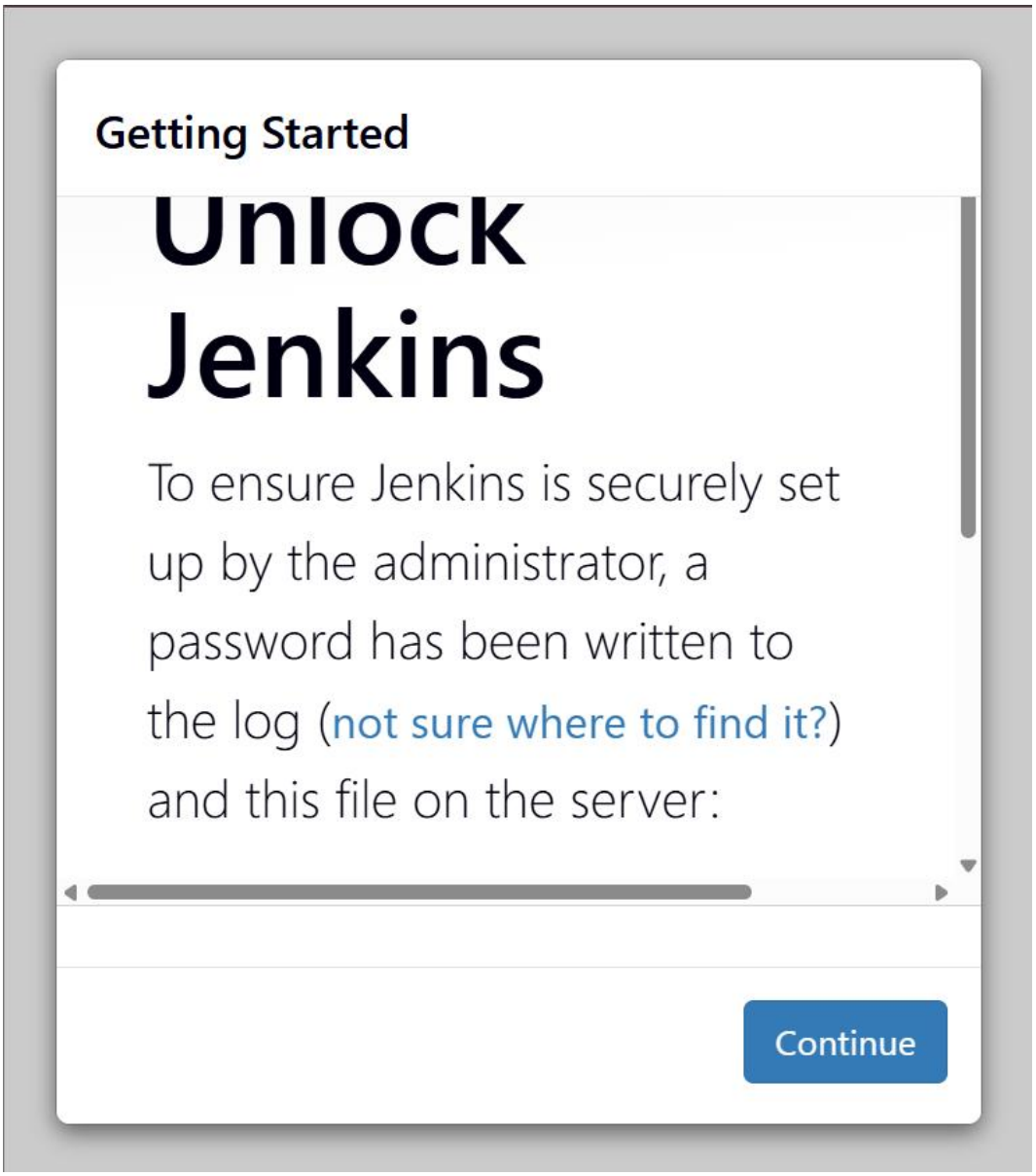
4. Verify Jenkins started

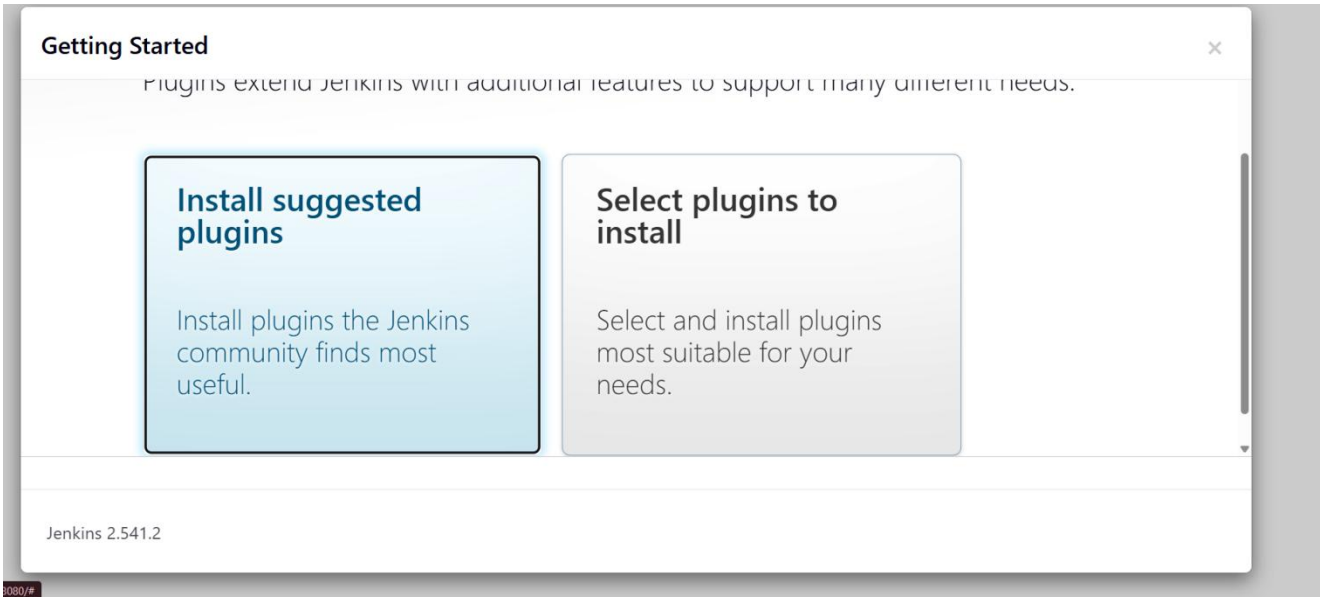
`docker ps`

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$ docker ps
CONTAINER ID   IMAGE            COMMAND                  CREATED         STATUS         PORT
S                                                                                          N
AMES
c795e0e45c82   jenkins-docker   "/usr/bin/tini -- /u…"   2 minutes ago   Up 2 minutes   0.0.
0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   j
enkins
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$
```

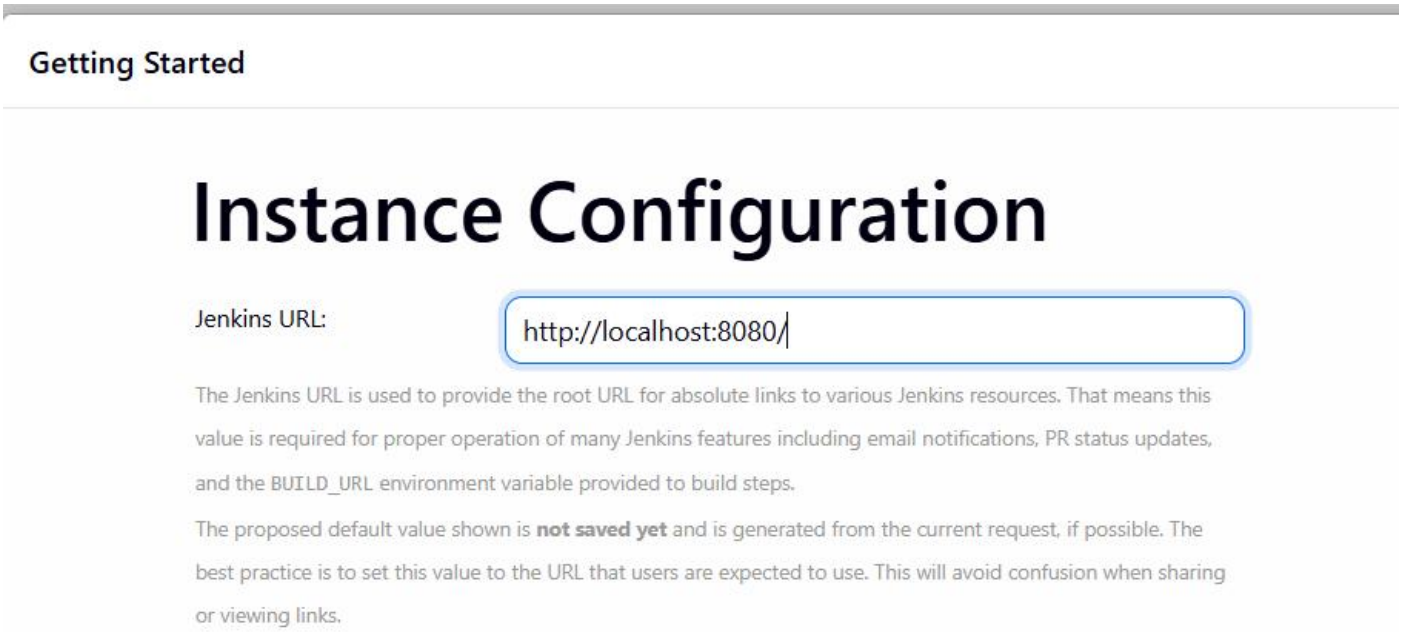
5. Open Jenkins

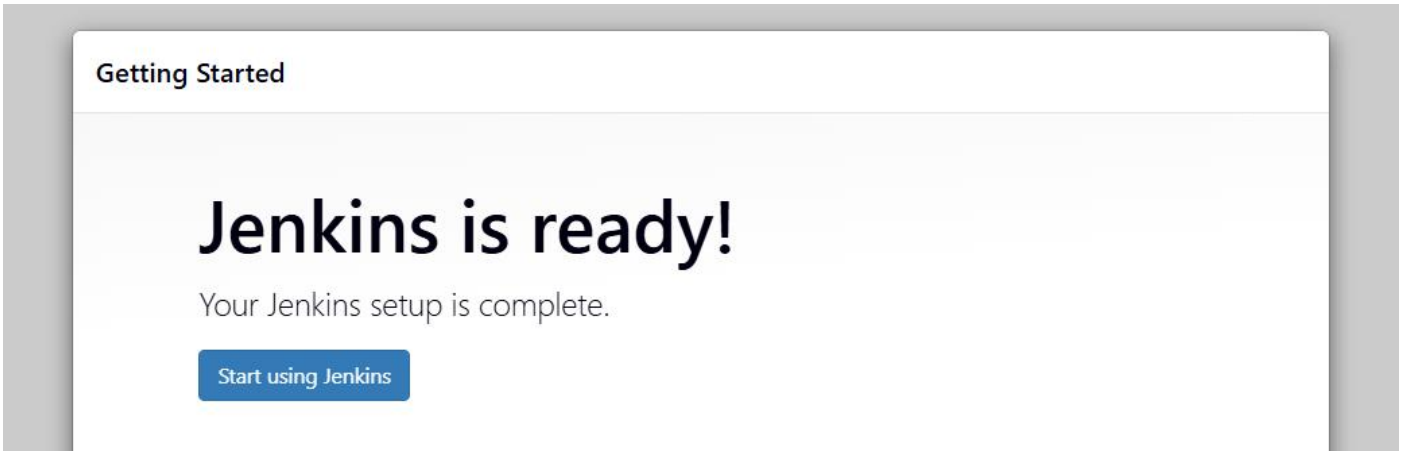
`http://localhost:8080`

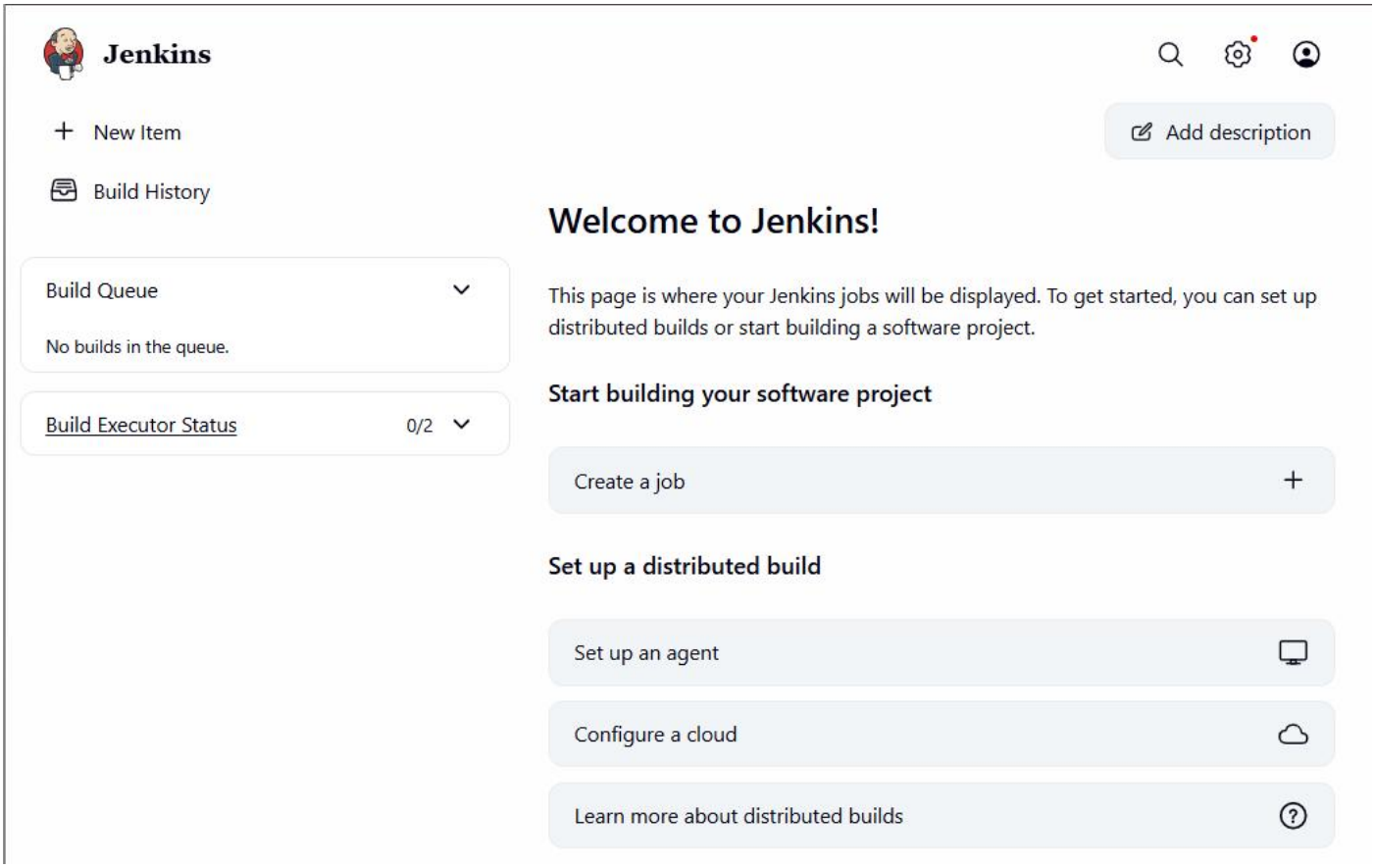




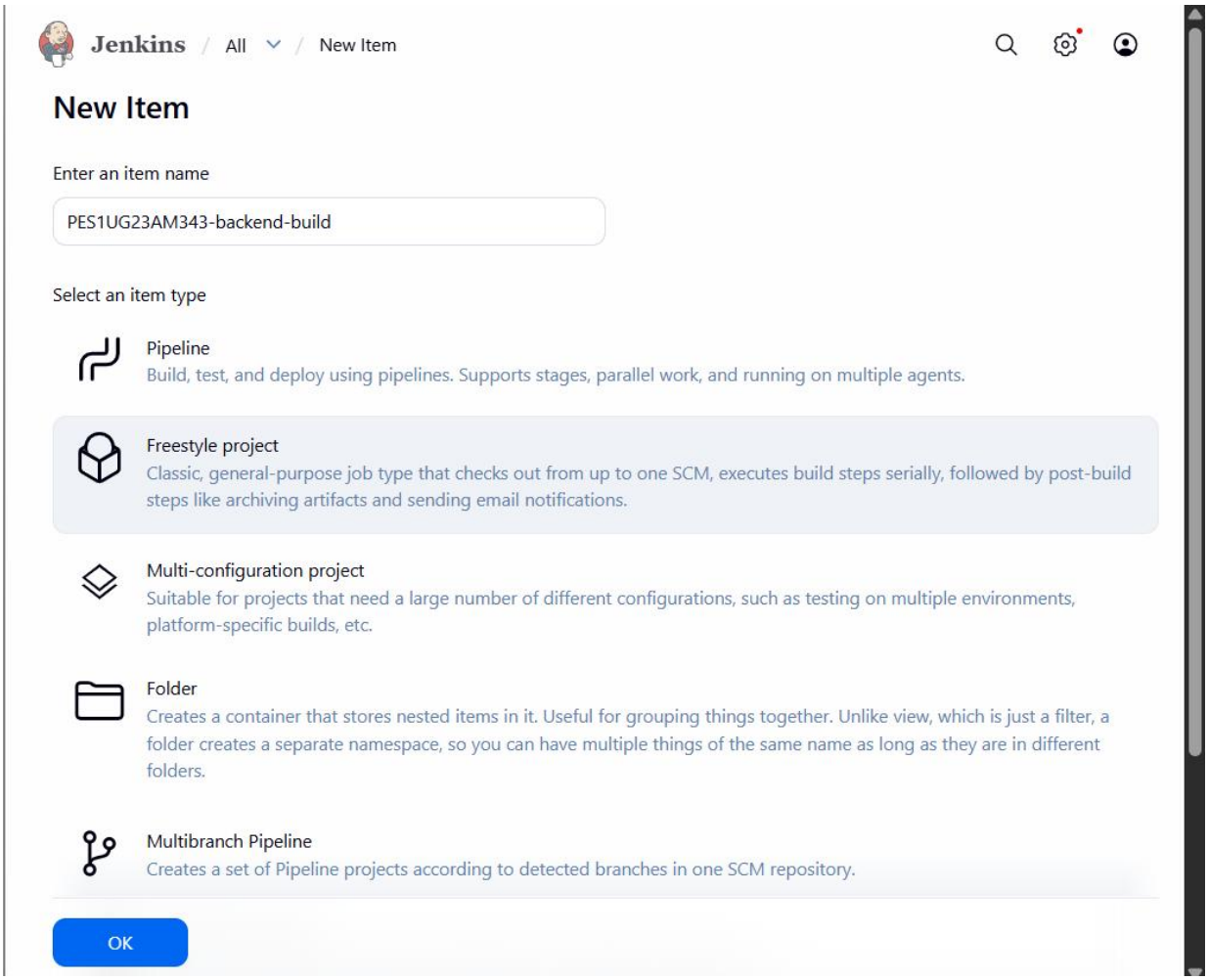
Install all necessary dependencies







Create a new project using Jenkins



#10 (Mar 9, 2026, 9:02:17 AM)

Started by user Vardha Kathuria

This run spent:

- 5 ms waiting;
- 2 min 3 sec build duration;
- 2 min 3 sec total from scheduled to completion.

Revision: d3c28a04113b3869c158db6db5ba156e4afbb7d6
Repository: https://github.com/voxility2010/PES1UG23AM343_CC_LAB

- refs/remotes/origin/main

No changes.

```
#10 exporting to image
#10 exporting layers
#10 exporting layers 22.5s done
#10 exporting manifest sha256:718792d8717f21229b93da67510cbecb559f80262a39c2f9a86c800b9ae4a412 0.0s done
#10 exporting config sha256:7fea8d9692a05792693dbd81ae2af5044f338ab07a6579c5f5e089ca4526cf1f 0.0s done
#10 exporting attestation manifest sha256:42bfa96b50840aa3183a592e676a317320af44d4d574aa76a80c911c323563e4
#10 exporting attestation manifest sha256:42bfa96b50840aa3183a592e676a317320af44d4d574aa76a80c911c323563e4 0.1s done
#10 exporting manifest list sha256:9ece07cdf7032d15c6078e3e2344cf89d3f2023cac8b0f2cbd0458aeb075a35e 0.0s done
#10 naming to docker.io/library/backend-app:latest done
#10 unpacking to docker.io/library/backend-app:latest
#10 unpacking to docker.io/library/backend-app:latest 8.0s done
#10 DONE 30.7s
Finished: SUCCESS
```

Build with Parameters:

```
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$ docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS          PORTS
                                  NAMES
4eec681387cf   backend-app           "./app"                  8 seconds ago    Up 8 seconds    8080/tcp
                                  backend2
941ae2458afc   backend-app           "./app"                  9 seconds ago    Up 9 seconds    8080/tcp
                                  backend1
d86fb5d65c0b   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   24 minutes ago   Up 24 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->5
0000/tcp, [::]:50000->50000/tcp   jenkins
PES1UG23AM343:/mnt/c/Users/acer/OneDrive/Documents/VARDHA/SEM 6/CC_LAB-6/CC_LAB-6$
```

#5 (Mar 9, 2026, 9:28:16 AM)

Started by user Vardha Kathuria

This run spent:

- 13 ms waiting;
- 15 sec build duration;
- 15 sec total from scheduled to completion.

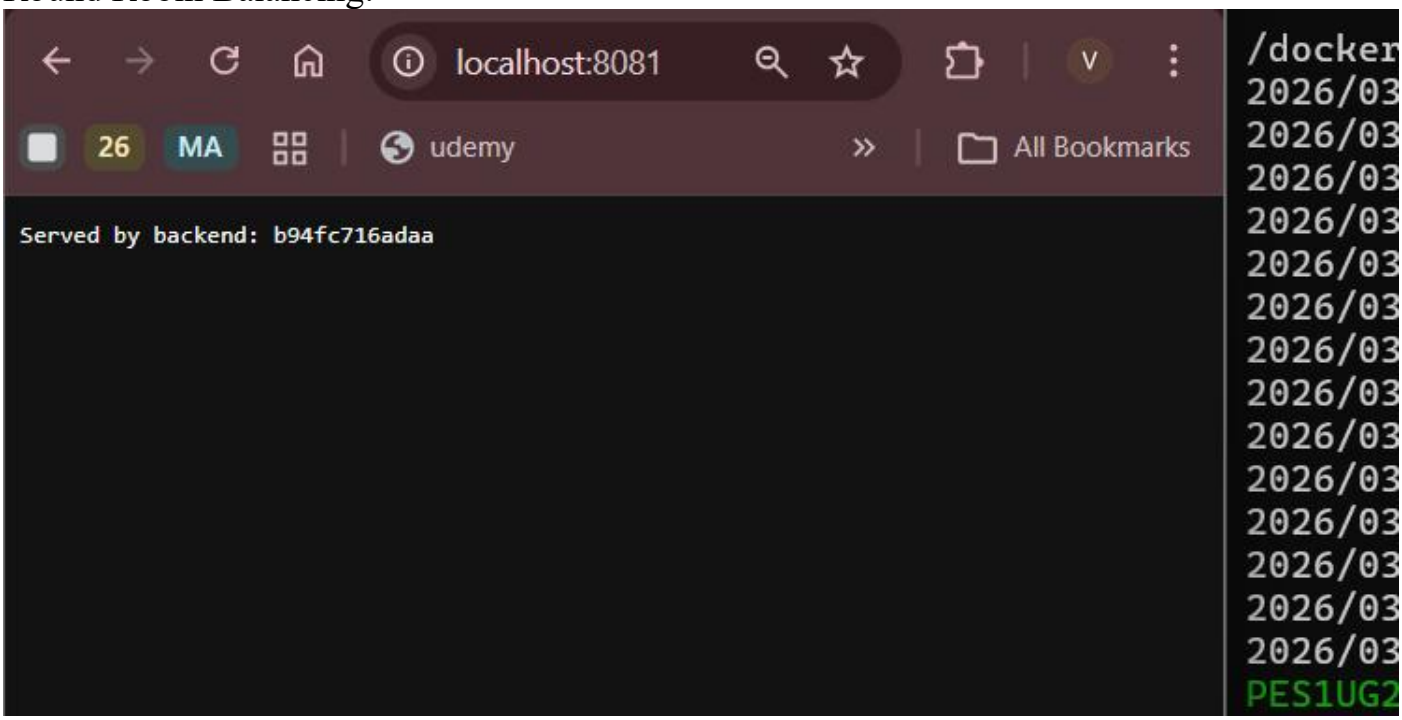
Revision: 48e16a9947bc8432dbb26b3dc08c4fd5c744637e
Repository: https://github.com/voxility2010/PES1UG23AM343_CC_LAB

- refs/remotes/origin/main

Update Jenkinsfile for backend and NGINX deployment 48e16a9
noreply at 9:28 AM 3/9/26

```
35/ae4df6c06/bf997e67/0f9b087c3fc091dd/1472 5f22b31a2044c450f2ef2f
+ docker run -d --name backend2 --network app-network backend-app
26651c225a0505c98c9d9fe16738d5ee9ee786d7184b23008d263054a151b7cd
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy NGINX Load Balancer)
[Pipeline] sh
+ docker rm -f nginx-lb
nginx-lb
+ docker run -d --name nginx-lb --network app-network -p 8081:80 nginx
807ba87e599117eac34b64db3e594281cfdccbcb6fe4ef309f15d717403d1c70
+ docker cp LAB_6/nginx/default.conf nginx-lb:/etc/nginx/conf.d/default.conf
+ docker exec nginx-lb nginx -s reload
2026/03/09 09:28:31 [notice] 37#37: signal process started
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage (hide)
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully. NGINX load balancer is running.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Round Robin Balancing:



Least Connections

```
upstream backend_servers {
    least_conn;
    server backend1:8080;
    server backend2:8080;
}

server {
    listen 80;

    location / {
        proxy_pass http://backend_servers;
    }
}
```

LAB6-PIPELINE-NGINX

Permalinks

- Last build (#9), 1 min 20 sec ago
- Last stable build (#9), 1 min 20 sec ago
- Last successful build (#9), 1 min 20 sec ago
- Last failed build (#6), 17 min ago
- Last unsuccessful build (#6), 17 min ago
- Last completed build (#9), 1 min 20 sec ago

```
+ docker run -d --name backend2 --network app-network backend-app
626a1fa0faef2b35ae633c814678021cbfad02da47cc72a29452197816d473ef
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy NGINX Load Balancer)
[Pipeline] sh
+ docker rm -f nginx-lb
nginx-lb
+ docker run -d --name nginx-lb --network app-network -p 8081:80 nginx
6aed2447425737fbef4a6b26c4d02eaf2e130e7694f20a8d2e3a2c3b7f53b115
+ sleep 3
+ docker cp LAB_6/nginx/default.conf nginx-lb:/etc/nginx/conf.d/default.conf
+ docker restart nginx-lb
nginx-lb
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully. NGINX load balancer is running.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```